Akoizo Original Practice - Form 1 - Answer key

Division B | Meteorology | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Which satellite channel can show a jet stream in clear air?

Unit: Observing severe weather: stations, balloons, radar, and satellites.
Worked-case data: A radar image shows a storm with reflectivity of 65 dBZ, a tight red-green velocity couplet on its southwest side, and a small area of CC values around 0.7 at the same spot. What is happening?
Separate investigation observations:
Reflectivity: A hook-shaped appendix on the storm's southwest side.
Velocity: Strong inbound and outbound winds side by side at the hook.
Correlation coefficient: A small area of 0.6-0.7 inside the hook.

A. Water vapor

B. Visible

C. Infrared only at night

D. Radar reflectivity

Correct: A

2. (2 points) Unsaturated air rises and cools at about:

Unit: Stability on Stüve and Skew-T diagrams.
Worked-case data: The surface temperature is 30 °C and the dew point is 22 °C. Estimate the LCL and comment on tornado potential.
Separate investigation observations:
Surface: Temperature 32 °C, dew point 23 °C.
Layer near 850 mb: A thin warm layer: temperature rises with height.
Above 700 mb: Temperature drops steeply with height.
Afternoon: A front lifts the air by midafternoon.

A. 20 °C per km

B. 9.8 °C per km

C. 2 °C per km

D. 6.5 °C per km always

Correct: B

3. (2 points) Which feature defines a supercell?

Unit: Thunderstorms and lightning.
Worked-case data: You see a flash and hear thunder 12 seconds later. How far away is the lightning, and are you safe outdoors?
Separate investigation observations:
Duration: The storm has lasted three hours.
Radar: A hook echo and a bounded weak-echo region.
Velocity: Persistent rotation in the mid levels.
Reports: Hail up to 2.75 inches (baseball size).

A. A single short-lived cell

B. A warm front

C. A rotating updraft (mesocyclone)

D. A gust front

Correct: C

4. (2 points) A hurricane with sustained winds of 115 mph is:

A. Category 2

B. Category 4

C. Category 1

D. Category 3

Correct: D

5. (2 points) On Doppler velocity, green colors usually mean:

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A. Motion away from the radar

B. Heavy rain

C. Hail

D. Motion toward the radar

Correct: D

6. (2 points) On a sounding, temperature and dew point lines close together indicate:

A. Nearly saturated air

B. Very dry air

C. An inversion

D. Strong wind

Correct: A

7. (2 points) In the dissipating stage of a single-cell storm:

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Radar: A hook echo and a bounded weak-echo region.
Velocity: Persistent rotation in the mid levels.
Reports: Hail up to 2.75 inches (baseball size).

A. Rain has not begun

B. The downdraft dominates

C. The updraft strengthens

D. The cloud is still growing

Correct: B

8. (2 points) Where are a hurricane’s strongest winds?

Unit: Hurricanes, typhoons, and cyclones.
Worked-case data: A hurricane with 140 mph sustained winds approaches a coast with a wide, shallow continental shelf at high tide. Rate the storm and assess surge.
Separate investigation observations:
Satellite: A clear eye surrounded by a ring of very cold cloud tops.
Recon aircraft: Sustained surface winds of 160 mph.
Motion: Slowing to 5 mph near the coast.
Coast: A wide, shallow bay ahead of the storm.

A. In the outer rain bands

B. Far ahead of the storm

C. In the eyewall

D. In the eye

Correct: C

9. (2 points) Why are cold, bright cloud tops on infrared images important?

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A. They show surface temperature

B. They indicate clear skies

C. They mark tall thunderstorms

D. They mark fog

Correct: C

10. (2 points) An environmental lapse rate between the moist and dry rates is:

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A. Absolutely stable

B. Absolutely unstable

C. Always neutral

D. Conditionally unstable

Correct: D

11. (2 points) In the usual charge structure, the upper part of a thunderstorm is:

A. Positively charged

B. Negatively charged

C. Neutral

D. Charged only during rain

Correct: A

12. (2 points) How does El Niño usually affect Atlantic hurricane activity?

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Separate investigation observations:
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Recon aircraft: Sustained surface winds of 160 mph.
Motion: Slowing to 5 mph near the coast.
Coast: A wide, shallow bay ahead of the storm.

A. It warms the Atlantic and doubles storms

B. It reduces activity by increasing shear

C. It increases activity

D. It has no effect

Correct: B

13. (4 points) Explain why mesonets help forecasters during severe weather.

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answer: Mesonet stations are much closer together than airport ASOS stations, so they detect small boundaries such as outflows and drylines where storms often form. (4 points).

14. (4 points) Explain why saturated air cools more slowly as it rises.

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answer: Condensation releases latent heat, which partly offsets the cooling from expansion. (4 points).

15. (4 points) Explain how a multicell storm keeps producing new cells.

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answer: Cold outflow from mature cells spreads out as a gust front that lifts warm, moist air, starting new updrafts along the flank. (4 points).

16. (4 points) List the conditions needed for a tropical cyclone to form.

Unit: Hurricanes, typhoons, and cyclones.
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Separate investigation observations:
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Recon aircraft: Sustained surface winds of 160 mph.
Motion: Slowing to 5 mph near the coast.
Coast: A wide, shallow bay ahead of the storm.

answer: Warm ocean water of about 26.5 °C to a good depth, moist air, low wind shear, a pre-existing disturbance, and a location at least about 5° from the equator. (4 points).

17. (4 points) How does composite reflectivity differ from base reflectivity?

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answer: Base reflectivity shows the lowest radar scan. Composite reflectivity shows the strongest value from any elevation above each point, so it can reveal hail cores aloft. (4 points).

18. (4 points) Describe how a capping inversion affects storm development.

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answer: Warm air aloft stops parcels from rising, suppressing storms. If heating or lifting breaks the cap, the stored instability can be released suddenly, producing strong storms. (4 points).